
Deep Reinforcement Learning for Portfolio Management

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Abstract

Modern Portfolio Theory (MPT) and its Mean-Variance Optimization (MVO) framework have long guided financial portfolio allocation but rely on assumptions that may not hold in real markets. Deep Reinforcement Learning (DRL) has demonstrated improved backtested performance over MVO but is often evaluated without accounting for practical constraints like transaction costs and slippage [1]. Additionally, DRL is rarely compared against passive strategies such as dollar-cost averaging (DCA). This study extends prior research by incorporating these real-world factors to assess whether DRL can consistently outperform MVO and DCA in risk-adjusted returns. Using sector-based market data, volatility indicators, and estimated trading costs, we implement a DRL model trained via Proximal Policy Optimization (PPO) to dynamically adjust portfolio weights. This project provides insights into the viability of DRL for real-world portfolio management.

1 Problem and goal

1.1 Objective

For the past 70 years, Modern Portfolio Theory (MPT) has been widely used in financial portfolio optimization. It balances risk and return using historical covariance estimates and expected returns. However, traditional MPT-based approaches, particularly Mean-Variance Optimization (MVO), rely on assumptions such as normally distributed returns and stable correlations, which may not hold in real-world markets. Deep Reinforcement Learning (DRL) has shown promise as an alternative approach for portfolio optimization, as demonstrated in "Deep Reinforcement Learning for Optimal Portfolio Allocation: A Comparative Study with Mean-Variance Optimization [1]." However, this study, and other existing studies often fail to account for practical challenges such as slippage and trading costs, which can significantly impact the effectiveness of portfolio strategies. Additionally, comparisons are typically limited to MVO and DRL, without considering simpler passive investment strategies such as dollar-cost averaging (DCA) in broad market indices like the S&P 500. This project aims to extend the research performed by Sood et al. [1], by incorporating these real-world factors and evaluating whether DRL can consistently outperform both MVO and DCA in terms of risk-adjusted returns (Sharpe ratio).

1.2 Relevance

While DRL has demonstrated superior backtested performance over MVO in previous studies, its real-world applicability remains uncertain due to the exclusion of transaction costs and slippage, both of which can erode returns, particularly for high-frequency strategies. Additionally, investors often compare active strategies not only against MVO but also against simple passive investment approaches like DCA. If DRL-based portfolio allocation can outperform both traditional optimization techniques and passive investment strategies while accounting for real-world costs, it could provide a compelling case for its adoption in portfolio management. This research is

important for financial professionals, institutional investors, and quantitative researchers looking to integrate machine learning into investment decision-making. Moreover, I'm curious about exploring another avenue to balance my personal portfolio.

1.3 Expected Results

I expect that DRL will be competitive with MVO and maybe perform better in terms of the Sharpe ratio and lower maximum drawdowns. However, when factoring in real-world considerations such as slippage and trading costs, I believe the differences between the two strategies will be much smaller – if not DRL will perform worse. Furthermore, I expect both DRL and MVO are likely to underperform compared to a simple dollar-cost averaging (DCA) strategy in the S&P 500.

2 Model architecture

2.1 Data

The dataset for this study will be collected using `yfinance*`, an open-source Python tool for retrieving financial data. The dataset includes:

- **S&P 500 Sector Data:** Daily OHLCV (Open, High, Low, Close, Volume) data for the 11 sectors.
- **Market Volatility Indicators:** VIX, rolling standard deviations (e.g., 20-day and 60-day volatility (derived from market data)).
- **Trading Costs and Slippage Estimates:** Derived from bid-ask spreads and liquidity conditions. OR, modeled using a flat transaction fee (percentage of transaction).

*Potentially may use Yahoo finance's official, paid API – would prefer not to.

2.2 Deep learning formulation

This project will utilize Deep Reinforcement Learning (DRL) to solve the portfolio optimization problem. The DRL agent's state space consists of 60-day log returns for each sector (may change how many days), portfolio composition at the previous timestep, market volatility indicators (such as VIX and rolling standard deviations), cash position in the portfolio, and estimated transaction costs for rebalancing. These inputs provide a comprehensive representation of the market environment, allowing the agent to make informed trading decisions.

The action space is defined as the allocation of portfolio weights across the 11 sectors, constrained such that the total allocation sums to one and remains within a valid range for long-only positions. The agent adjusts these allocations dynamically to optimize performance over time.

The reward function is centered around maximizing risk-adjusted returns, specifically using the Sharpe ratio as a key performance measure. Penalty terms are incorporated to account for transaction costs and slippage.

For learning, this project will utilize Proximal Policy Optimization (PPO), a model-free RL algorithm known for stable convergence in policy gradient methods. The training environment is structured as a market replay simulation, enabling the agent to learn from historical price movements and refine its decision-making processes in a realistic setting.

3 Schedule

Table 1: Schedule

Due date	Deliverables
2/29	Collect and clean data. Also calculate relevant metrics (rolling standard deviations, etc.), and determine method for modeling transaction fees.
3/16	Complete PPO Training
3/21	Complete preliminary report
4/13	Complete PPO backtesting

4/15	Complete MVO and DCA returns and backtesting
4/24	Complete presentation
5/9	Complete final report

References

[1] Sood, S., et al. "Deep Reinforcement Learning for Optimal Portfolio Allocation: A Comparative Study with Mean-Variance Optimization." *Proceedings of the FinPlan 2023 Workshop at ICAPS 2023*, 2023, <https://icaps23.icaps-conference.org/>.